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Caught them all: Gaming disorder, motivations for playing and spending among core Pokémon Go players

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ABSTRACT

Pokémon Go (2016) not only enjoys tremendous worldwide appeal among players, the game has also captivated the scientific community. The aim of this study was to examine the characteristics and motivations of hardcore Pokémon Go players by analyzing self-reported data from 1630 trainers of level 25 and higher. There were more female players in our sample, yet males played more and showed more progress in the game. In general, the need for in-game relatedness was the strongest predictor of playing, followed by the need for in-game competence. Male players showed stronger in-game motivational needs for relatedness and competence, whereas female players were more strongly motivated by a need for autonomy. The needs for relatedness and competence predicted spending on microtransactions, whereas the need for autonomy did not. Depending on the measure for gaming disorder, between 1% and 9% of these core players met the diagnostic criteria. Gaming disorder was also related to social anxiety and increased spending on in-game microtransactions.

1. Introduction

In the summer of 2016, millions of people across the world were walking around streets and parks, frantically flicking Pokéballs at pocket monsters on their smartphones. Thousands of players stampeded through the streets of Taipei, hoping to catch an elusive *Snorlax* [27] or swarmed Central Park for an ultra-rare *Vaporeon* [28]. With more than 500 million downloads in the first few months alone, Pokémon Go quickly became a global phenomenon [23]. Although it seemed that the initial hype surrounding this location-based, augmented reality game had died by the time winter came along, the game's developer Niantic has been steadily introducing new elements, such as cooperative raids (June 2017), the ability to battle, befriend and trade with other trainers (June 2018), and seasonal *Battle Leagues* (November 2020). In addition to increasing the available Pokémon from 150 to 500 over the years, these social and competitive additions seem to have increased the longevity of the game considerably, with reportedly more than 140 million steady players and revenues of approximately $800 million every year since its release [3].

The game not only enjoys tremendous global popularity among players, the game has also captivated the scientific community. With more than 100 published studies, Pokémon Go has quickly become the most-researched video game. The majority of these studies focused on the potential physical benefits of walking outside while playing [1;45]. Others have focused on the enjoyment of the game and its positive effects on emotional well-being [5]. Some studies have focused on the motivations of players [4], and a few studies have examined excessive or problematic use of Pokémon Go [9]. In general, research shows that Pokémon Go has a positive effect on players' physical, mental, and social health (for an overview see [61]).

Despite the abundance of research on this game, persistent, long-term players have not been examined specifically. These persistent players have been labeled 'hardcore', a term used by gamers themselves to express the strongest possible identification with gaming [66]. Core gamers are comprised of diverse subpopulations with distinct genre preferences [58], including a group with a strong preference for Pokémon Go. Understanding the long-term motivations and spending patterns of core players is not only interesting from an academic perspective, free-to-play mobile games such as Pokémon Go have tremendous economic relevance because their microtransaction-monetization model has become the most profitable within the game industry [48]. However, the success of Pokémon Go has also caused some concern among some researchers, who suggested that players' obsession with its collection mechanics can lead to game addiction [52]. Considering recent efforts of leading mental health organizations to legitimize excessive, obsessive and problematic gaming as gaming

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disorder [2,62], we examined how different measures of this maladaptive behavior are related to the characteristics such as social anxiety [38], spending patterns [13], and motivational needs satisfaction [47] of core Pokémon Go players.

Within research on the motivations of players, the *Self Determination Theory* (SDT) is one of the most widely used approaches to understanding which inherent human needs are satisfied through games [57]. Previous studies have reported gender differences in autonomy and competence such that women generally felt more autonomous and less competent than men [11,40]. These perceived differences motivate male and female players to seek different compensatory need satisfaction in games. For instance, male players reported a greater need for autonomy during gameplay than female players did [12]. Male players were also more motivated by the need to experience a sense of relatedness [44]. These motivational gender differences may also affect the playing and spending habits of Pokémon Go players.

The overall aim of the current study was to examine the motivational pull of Pokémon Go that has been keeping millions of players engaged around the world for years. Who are the core players that make this game a popular and financial success? The aims of this study were threefold: 1) to determine the general characteristics, motivations and spending patterns of its core players; 2) to examine the moderating role of gender within the effects of the motivational need for in-game competence, autonomy and relatedness on playing- and spending patterns; and 3) to examine how signs of gaming disorder are related to player characteristics, motivations, playing- and spending patterns. To meet these aims, a survey study was held among more than 1600 core Pokémon Go players.

2. Theory

2.1. Player motivations

It is commonly understood that people actively select and use media entertainment to satisfy their needs [56]. Within research on video games, the *Self Determination Theory* (SDT) is one of the most widely used approaches to understanding player motivations [57,66]. According to the SDT, three intrinsic human needs motivate users to engage with video games: 1) *Competence*, the need to control the outcome and experience mastery, 2) *Autonomy*, an individual's perceived volition or control; 3) *Relatedness*, the need to experience interaction and connection with others [11]. Studies have shown that satisfaction of all three needs is associated with greater game enjoyment [53], increased time spent playing, and problematic gaming [47].

In general, competence plays a crucial role in motivating gamers. Competence is experienced when players improve and expand their skills by overcoming challenges, which include completing specific tasks, or competing against other players [67,68,69]. In Pokémon Go, challenges can include, gaining a level, collecting rare Pokémon, gaining experience points, gathering virtual game objects, or competing with others for dominance over a gym [15]. A study among college students indicated that catching Pokémon was the strongest predictor of game enjoyment [45]. Similar findings came from a study by [70] as indicated by players' strong desire to catch all Pokémon, achieve the highest level, and compete with other players. When people experience these forms of achievement, they display stronger intentions to continue playing Pokémon Go [7] and are more likely to perceive the game as more interesting [71]. A recent study that included SDT motivations concluded that players' need for competence was the strongest predictor of intention to play [35]. Thus, the experience of in-game competence is considered a prominent motivation for core players of Pokémon Go.

The need for autonomy describes our desire to regulate and control our own behavior [73]. In games, opportunities for choice and non-controlling instructions have shown to enhance players' sense of autonomy [74]. Games that provide character customization options can also enhance the sense of autonomy among players [50]. Customization options provide players a sense of volition and control by allowing them to change the attributes and appearance of their game character [16]. Studies have shown that aesthetic avatar customization options resulted in greater game enjoyment [21,32]. Customization also contributed to loyalty among players [59]. Although some researchers found that the need for in-game self-presentation did not predict continuance intention [7], a recent study on Pokémon Go indicated that customization of avatars provides players with the experience of autonomy, which positively influenced both game enjoyment, and time spent playing [75]. Because there is substantial evidence that players' need for autonomy may predict intention to play Pokémon Go [35], similar needs for in-game autonomy through customization are expected among the core players in the current study.

Studies have repeatedly shown that social interaction is a key factor in motivating players' continued engagement in online games [8,41,76,77]. The sense of being part of a community enhances loyalty and has a positive effect on the intention to continue playing online games [15]. Similarly, players of Pokémon Go have highlighted the need for relatedness when they indicated that social benefits drive its continued use [54]. Furthermore, when people experience social interaction, they display stronger intentions to continue playing Pokémon Go [7,14]. Moreover, studies have shown that playing Pokémon Go facilitated friendship initiation and intensification [5], increased a sense of social belonging [60], and provided a strong sense of community [64]. Thus, relatedness-supportive elements in Pokémon Go seem particularly relevant to motivating its core players.

Motivations for playing Pokémon Go may differ between male and female players, mirroring the general gender differences in preference for game genres and content. For instance, male players generally show a stronger preference for competition in games [20,42], indicating that male players likely show a stronger motivational need for competence. Among players of Pokémon Go, compared to female players, male players not only seek more achievement, but also more social interactivity. This study further showed that men play Pokémon Go mainly for social interactivity and achievement, whereas women mostly play for enjoyment gratifications [44]. Regarding the need for in-game autonomy, adolescent and young adult males scored higher than females on all motivation scales, except the desire for customization [43]. Therefore, gender differences in their desire for in-game motivational needs are expected to influence their use of Pokémon Go.

2.2. Excessive, obsessive, and problematic gaming

In 2013, the American Psychiatric Association included Internet gaming disorder as a tentative disorder in the appendix of the DSM-5 [2]. Five years later, Gaming Disorder was included in the World Health Organization's [85] 11th International Classification of Diseases [62]. The world's two leading mental health organizations thereby agree that (Internet) Gaming Disorder may pose a severe health risk to excessive game players around the world. Nevertheless, these two organizations show striking differences in their approach to formally diagnose this disorder. The DSM-5 proposes nine criteria for IGD, with a recommended threshold of experiencing five or more criteria in the last twelve months for a positive diagnosis. In contrast, the ICD-11 provides three clearly demarked criteria that describe gaming disorder as: 1) a pattern of gaming behavior characterized by impaired control over gaming; 2) increasing priority given to gaming over other activities, and; 3) continuation of gaming despite the occurrence of negative consequences.

Although there is no scientific evidence of (Internet) gaming disorder among players of Pokémon Go, some researchers have warned that "users should be made aware of the particularly addictive nature of the Pokémon GO" [52]. Similarly, the alleged addictive nature of this game was also mentioned in research by Broom et al. [4]. Non-players indicated that one of the main reasons for *not* wanting to start playing Pokémon Go, was its presumed addictive nature. One of the few studies

that measured 'addiction behavior of Pokémon Go', indicated relatively low levels of game addiction among players in Malaysia [25]. In the current sample of core players, signs of gaming disorder or IGD may differ, depending on the measurement.

In general, male players generally show a greater tendency toward game addiction than female players do [38,78]. Furthermore, several studies have shown that players who experience social anxiety may be more susceptible to game addiction. Social anxiety can be defined as experiencing feelings of nervousness in social situations [78] and has been shown to be an antecedent of game addiction [39]. Similarly, problematic MMORPG players tend to have a socially anxious profile [80]. Studies on socially anxious players of Pokémon Go indicated they tended to catch less Pokémon and gained less experience points during gameplay [30]. Furthermore, socially anxious players experienced less well-being as a result of playing [5]. Thus, we expect socially anxious core players to show more signs of gaming disorder or IGD.

2.3. Microtransactions

Free-to-play (or freemium) games refer to types of games that can be acquired and played for free, with revenue being generated through sales of in-game products and features that have functional or cosmetic value for its players [18,101,102]. Over the past decade, the game industry's overall revenue model has shifted its focus from selling games to selling optional features that can be acquired through microtransactions in online games [82]. The monetization strategy is to entice players to make multiple small financial transactions (i.e., microtransactions) for optional features that may enhance their game performance or optimize their overall gameplay experience [34], [83]. Research has shown that player retention is essential for the financial success of free-to-play games. Players who developed a sense of loyalty to a game showed increased willingness to purchase virtual goods [17,24]. Pokémon Go is also free-to-play and its success largely depends on persistent players willingness to make microtransactions to enhance their gameplay experience. A recent study on in-game purchases among Pokémon Go players in Taiwan found that 38.4 % of their sample had spent money on the game [84]. Among core players, spending is perhaps even more common.

Several studies have examined the relations between motivations and spending on microtransactions in Pokémon Go. By placing limitations on resources and inventory space, the need for competence can drive players to spend money to overcome these restrictions. Specifically, online gamers who sought gratification related to competition were more likely to purchase in-game content [19]. Furthermore, players were more likely to buy items that provided them with a greater sense of authenticity [63] or items that allowed for stronger identification with their characters [49]. Thus, players' sense of authenticity and identification can be enhanced through purchased character customization options, fulfilling the need for in-game autonomy. Regarding relatedness, socializing with other Pokémon Go players has been associated with in-app purchase intentions [19]. Similarly, a stronger sense of involvement in the community predicted in-game purchase intentions [15]. Finally, a recent study found that when controlling for age, gender, and income, only players who were motivated by relatedness (not autonomy or competence) were more likely to make in-game purchases [35]. In short, all SDT motivations may drive spending on microtransactions in Pokémon Go.

Excessive, obsessive and problematic gaming may also influence in-game purchases. For instance, Dreier et al. [13] found a positive association between in-game spending patterns and addiction to online games. The group with the biggest spenders (i.e., *Whales*) also had the most addicted gamers, whereas the group of so-called *Freeloaders* had the least. Conversely, a study on microtransactions in Fortnite [34] did not find a significant relationship between spending and IGD. Nevertheless, a recent literature review on microtransactions and gaming disorder concluded that there was clear evidence of positive relationships between microtransactions and IGD [55]. Interestingly, this conclusion was based mostly on the relation that was found between problematic gaming and *loot box* spending [6]. Loot boxes are packs, chests, or boxes, containing a selection of random items (i.e., loot) that may enhance the gameplay experience. According to King and Delfabbro [33,99] loot boxes are "predatory monetization schemes" (p. 1967) that contribute to potential financial harm for those with Internet gaming disorder. However, Pokémon Go does not apply loot box mechanics, and to date there is no study on the relation between spending on Pokémon Go and signs of gaming disorder. Therefore, the current study aims to explore this relation.

3. Method

3.1. Sample

A cross-sectional online questionnaire was distributed among Pokémon Go players. In total, 2131 respondents started the online questionnaire. However, a large group (n = 489) stopped responding after the initial usage items (<25 % of the survey) and therefore did not provide any information on the SDT motivations or other relevant variables. Therefore, these respondents were removed from the analysis. Another 12 respondents were removed for having relatively little experience with Pokémon Go. In the current study, core players were considered trainers who were level 25 or higher (N = 1630). Out of these 1630 core Pokémon Go players, the majority was female (n = 989, 60.7 %). 11 respondents indicated non-binary and 3 did not want to reveal their gender, leaving 627 male players (38.5 %). Age ranged from 16 (minimum age requirement for consent) to 75 years old (M = 36.8, SD = 11.62). The vast majority answered the Dutch version of the questionnaire (83.6 %, n = 1363). There were no significant differences between responses on the Dutch and English versions of the questionnaire on any of the relevant variables. Most of the respondents resided in the Netherlands (79.7 %, n = 1299), followed by Singapore (8.5 %, n = 139), Belgium (4.5 %, n = 74), and the USA (4.5 %, n = 73).

The majority of our sample (n = 1275, 78.2 %) had started playing Pokémon Go within the first two months after its release in Europe on July 16, 2016 (10 days after its Australian and North-American release). On average, the current sample of core gamers started playing 3.92 months after release (SD = 7.89). As indicated before, all 1630 respondents in our analyses had reached trainer level 25 or higher (M = 37.74, SD = 3.09), with almost half the sample (n = 796, 48.8 %) having reached level 40, the highest trainer level available at the time. Level progress in Pokémon Go requires exponential amounts of experience points (xp). For instance, the experience points required to get from level 25 to level 26 (190,000 xp) is <4 % of the amount of experience points required to get from level 39 to level 40 (5,000,000 xp). For each Pokémon caught, players receive between 100 and 1260 xp, depending on their throw style and the novelty of that Pokémon to their collection. Out of all players with a trainer level of 40, more than 78 % indicated they had caught more than 25,000 Pokémon, with 31 % indicating they had caught more than 50,000 Pokémon. Most players covered between 5 and 25 km per week while playing (n = 669, 41 %), with almost as many covering 25–50 km (n = 632, 39 %), and 16 % (n = 266) indicating they walked more than 50 km per week while playing Pokémon Go.

3.2. Procedure

In October 2019, an online questionnaire was distributed among players of Pokémon Go through Discord, an application designed for gaming communities that allows text, image, video and audio communication between users in a chat channel and through Facebook. The study was approved by the ethics commission from the department of Communication at the University of Amsterdam. Several international and regional Discord communities were contacted with requests to fill in the questionnaire. No compensation or prizes were awarded for

participation. The questionnaire consisted of approximately 90 items measuring player demographics, characteristics, motivations, game use and its appeal, microtransactions, and gaming disorder items. Specifically, we included age, gender, nationality, social anxiety, SDT-motivations, usage metrics, spending behavior, and gaming disorder. All included measures are further explained below. Respondents were asked to consult their Pokémon Go app for some of the metrics. Forced response was disabled for all items on the questionnaire. Completing the survey took about ten minutes after which respondents were thanked for their participation.

3.3. Measures

3.3.1. Usage

Player's use of Pokémon Go was measured in three ways: (1) Time spent per week; (2) Total amount of Pokémon caught; (3) Total experience points gained. First, players' weekly hours spent playing Pokémon Go (1) was calculated by multiplying the days per week each respondent indicated playing ($M = 6.31$, $SD = 1.52$) with the average number of hours each respondent played the game on a regular day ($M = 2.07$, $SD = 1.48$), resulting in the hours per week spent playing Pokémon Go ($M = 13.27$, $SD = 10.45$). The total number of Pokémon caught (2) was measured using 11 scaled answer-categories with increments of 5,000, ranging from '*< 5000*' through '*more than 50,000*'. The median was 5 (20,001 – 25,000, $M = 5.40$, $SD = 3.41$). The middle points of each category (opting for 2500 for the first, and 52,500 for the last category) were selected to get a clearer sense of the total number of Pokémon that were caught ($M = 24885.66$, $SD = 17475.87$). Finally, total experience earned in Pokémon (3) was measured by asking for their current trainer level (range 25–40, $M = 37.74$, $SD = 3.09$) and transposing that to the total experience points (in millions) required to reach that level (range 0.71 m – 20 m, $M = 14.01$ m, $SD = 6.63$ m).

3.3.2. SDT-motivations

According to the Self Determination Theory, three essential innate universal psychological needs motivate individuals to initiate behavior: the need for autonomy, competence and relatedness [11]. Items for the three scales were adapted from previous studies on SDT motivations in Pokémon Go [4,15]. *Competence* describes the need to control the outcome and experience mastery [11]. The need for competence in Pokémon Go was measured using four items, preceded by 'It is important for me to…". These four items were: 1) rise as quickly as possible in trainer level; 2) have better Pokémon than others; 3) catch as many unique Pokémon; 4) gather as many items as possible in Pokémon Go. Response categories ranged from 1 *(strongly disagree)* to 5 *(strongly agree)*. The items were averaged to create the scale scores. Cronbach's alpha for the competence scale was 0.73, indicating acceptable reliability ($M = 3.31$, $SD = 0.79$).

Autonomy describes the desire to be causal agents of one's own life and act in harmony with one's integrated self [11]. The need for autonomy in Pokémon Go was measured using four items, with response categories ranging from 1 *(strongly disagree)* to 5 *(strongly agree)*: 1) I enjoy collecting distinctive objects or clothing that have no functional value in the game; 2) It is important for me to customize my avatar; 3) I enjoy spending time customizing my avatar; 4) It is important to me that my avatar looks different from other avatars. The items were averaged to create the scale scores. Cronbach's alpha for this scale was 0.87, indicating good reliability ($M = 2.28$, $SD = 0.93$).

Relatedness describes the need to experience interaction and be connected to others [11]. The need for relatedness in Pokémon Go was measured using four items, with response categories ranging from 1 *(strongly disagree)* to 5 *(strongly agree)*. These four items were: 1) I enjoy playing Pokémon go because it allows me to meet other trainers; 2) I enjoy playing Pokémon go because it allows me to help other trainers; 3) I enjoy playing Pokémon Go because it allows me to get to know other players who I now consider my friends; 4) I enjoy playing Pokémon Go because it allows me to have meaningful conversations with other players. The items were averaged to create the scale scores. Cronbach's alpha for this scale was 0.90, indicating good reliability ($M = 3.29$, $SD = 1.01$).

3.3.3. Internet Gaming Disorder

The DSM-5 describes Internet Gaming Disorder (IGD) as persistent, recurrent and compulsive playing, to the exclusion of other interests, resulting in clinically significant impairment or distress [2]. IGD is present when a person meets five (or more) of the nine criteria (*Preoccupation, Tolerance, Withdrawal, Persistence, Escape, Problems, Deception, Escape, Conflict*) during a period of 12 months. Items were taken from the 9-item IGD scale [37]. Every item on the IGD scale was preceded by the statement: "During the last 12 months …", response options were either no (0) or yes (1). IGD scores were summed and ranged from 0 through 9. Exploratory Factor Analysis of the responses of 1531 respondents indicated that the scale was two-dimensional, with four items (*Conflict, Problems, Deception,* and *Persistence*) that loaded on a dimension indicating interpersonal friction and problems caused by excessive gaming. Nevertheless, the 9-item IGD sum-scale showed acceptable reliability with a Cronbach's alpha of 0.70 ($M = 1.60$, $SD = 1.82$). Out of 1531 players who responded to the IGD items, 134 (8.8 %) met at least five of the proposed diagnostic criteria.

3.3.4. Gaming disorder

As described in the 11th edition of the International Classification of Diseases (WHO, 2019), Gaming Disorder is characterized by a pattern of persistent or recurrent gaming behavior that involves 1) *Excessive behavior*: Impaired control over gaming (e.g., onset, frequency, intensity, duration, termination, context), 2) *Obsessive behavior*: Increasing priority given to gaming to the extent that gaming takes precedence over other life interests and daily activities, and 3) *Problematic behavior*: Continuation or escalation of gaming despite the occurrence of negative consequences. In the current study, these three concepts were consecutively measured using three selected items from the IGD scale [37]. Excessive behavior was measured with the item: 'Were you unable to reduce your time playing Pokémon Go, after others had repeatedly told you to play less?' Obsessive behavior was measured using the item: 'Have you lost interest in hobbies or other activities because Pokémon Go is all you wanted to do?' Problematic behavior was measured with the item: 'Have you experienced serious conflicts with family, friends or partner because of Pokémon Go?'. Out of 1531 responses, 15 players (1.0 %) indicated they had experienced each of these three criteria for gaming disorder over the last six months.

3.3.5. Social anxiety

Social anxiety was measured using six items from the *Social Interaction Anxiety Scale* [86]. This measure is focused on generalized social interaction anxieties (i.e., attending a social gathering, talking to acquaintances and strangers). Response categories ranging from 1 *(strongly disagree)* to 5 *(strongly agree)*. Sample items include: '*I have difficulty making eye contact with others'* and '*I tense up if I meet an acquaintance on the street*'. Cronbach's alpha for this scale was 0.90, indicating good reliability ($M = 1.83$, $SD = 0.80$).

3.3.6. Microtransactions

Based on a categorization of spending patterns by Dreier et al. [87], The amount of money players spent on Pokémon Go was divided into three categories. The first category consisted of players who had not spent any money on Pokémon Go, the *Freeloaders* ($n = 412$, 25 %), the second category consisted of players who had spent up to 50 euros on Pokémon Go, the *Dolphins* ($n = 615$, 38 %), and the top category consisted of the players who had spent more than 50 euros on Pokémon Go, the *Whales* ($n = 602$, 37 %). Mean spending category was 2.12 ($SD = 0.78$).

4. Results

4.1. Gender differences in usage and motivations

Table 1 shows the means of male and female players and their significant differences across relevant variables. In general, men showed more use of Pokémon Go; they had started playing almost two months earlier after its release, they spent approximately-two hours more on the game every week, they had acquired more experience points (i.e., higher trainer levels), and had caught about 6000 more Pokémon than women had. Men also showed more signs of gaming disorder, and spent more on microtransactions. Across all core players, the need for relatedness scored significantly higher ($M = 3.41$, $SD = 0.90$) than both the need for competence ($M = 3.26$, $SD = 0.71$, t (1622) = 6.31, p <.001) and the need for autonomy ($M = 2.28$, $SD = 0.93$, t (1622) = −38.47, p <.001). The need for competence was also significantly higher than the need for autonomy, t (1624) = 37.00, p <.001, indicating that the need for in-game relatedness is the most relevant motivation among core players, and the need for autonomy is the least relevant motivation. Interestingly, the need for in-game competence and relatedness was higher among male players, but the need for in-game autonomy was higher among female players. Cohen's effect sizes for gender differences ranged from small (Autonomy, $d = -0.15$, p =.003) to medium (Pokémon caught, $d = 0.37$, p <.001).

4.2. Relations between player characteristics, motivations and usage

Table 2 provides insight into the associations between player characteristics, the three motivational needs and the usage variables. These correlations indicate generally strong correlations between the needs for in-game competence and relatedness among players, their use of Pokémon Go, and the money spent on the game. Autonomy is the weakest predictor of weekly time spent on the game, and shows a weak negative association with the amount of Pokémon caught. Players' self-reported social anxiety was a negative predictor of experience acquired, and a positive predictor of the sum score of IGD criteria ($r = 0.22$, p <.001). Interestingly, social anxiety showed significant small negative correlations with experience points and total Pokémon caught, yet a significant positive correlation with weekly time spent playing. This may indicate that players with more social anxiety are less effective at progressing through the game. Finally, IGD was positively related to all usage criteria, including spending. Furthermore, the usage measures and spending were all interrelated (*r*s greater than 0.32, *p*s < 0.001), as were the three motivations (*r*s greater than 0.17, *p*s < 0.001). IGD sum scores showed positive correlations with all three motivations: Competence ($r = 0.21$, p <.001), Autonomy ($r = 0.15$, p <.001) and Relatedness ($r = 0.18$, p <.001).

Table 1
Gender differences in usage and motivations.

	Male Players ($n = 627$) *M* (*SD*)	Female Players ($n = 989$) *M* (*SD*)	T-test
Start Date	37.16 (7.66)	35.41 (9.63)	t (1533.48) = 4.05, p <.001, $d = 0.20$
Hours per week	14.68 (11.64)	12.35 (9.46)	t (1133.59) = 4.21, p <.001, $d = 0.22$
Experience points	15.30 (6.24)	13.21 (6.73)	t (1405.32) = 6.36, p <.001, $d = 0.32$
Pokémon caught	28.85 (17.58)	22.41 (16.93)	t (1284.46) = 7.25, p <.001, $d = 0.37$
Gaming disorder	1.83 (1.95)	1.45 (1.72)	t (1103.61) = 3.84, p <.001, $d = 0.21$
Spending	2.26 (0.77)	2.03 (0.77)	t (1336.14) = 5.89, p <.001, $d = 0.30$
Competence	3.41 (0.81)	3.25 (0.76)	t (1275.82) = 3.81, p <.001, $d = 0.20$
Autonomy	2.19 (0.95)	2.33 (0.91)	t (1614) = −2.96, p =.003, $d = -0.15$
Relatedness	3.41 (0.95)	3.23 (1.04)	t (1416.59) = 3.58, p <.001, $d = 0.18$

4.3. Motivational moderation models of gender and usage

Considering the numerous gender differences from Table 1, we tested whether gender moderates the effects of SDT motivations on usage, gaming disorder and spending. For that purpose twelve moderation analyses using 5,000 bootstrap samples were conducted with the PROCESS macro for SPSS (Hayes, 2012). The general model is shown in Fig. 1, and the corresponding unstandardized effect sizes, standard errors, and significance levels are shown in Table 3, along with significant model coefficients for each of the twelve moderation models. The first three models examined whether gender moderated the effect of the three SDT motivations on the total number of Pokémon caught by the players. The significant effect for gender indicates that male players, regardless of the need for in-game competence or autonomy caught more than female players did. Conversely, gender neither predicted nor moderated the effect of the need to experience relatedness as the strongest predictor of the number of Pokémon that were caught. In an effort to define what motivates players to catch them all, a linear regression model with all SDT motivations and gender as predictors of the amount of Pokémon caught was run, F (4, 1597) 57.38, p <.001, $R^2 = 0.13$. The need for relatedness was the strongest predictor ($b = 0.29$, $t = 12.18$, p <.001), followed by male gender ($b = 0.14$, $t = 6.00$, p <.001), and the need for competence ($b = 0.07$, $t = 2.74$, p =.006). Interestingly, the need for in-game autonomy (i.e., customization) became a significant negative predictor of the amount of Pokémon caught ($b = -0.09$, $t = -3.69$, p <.001).

Time spent playing Pokémon Go was significantly predicted by all three motivational needs. The higher the motivational needs for in-game competence, autonomy and relatedness, the more time was spent playing the game each week. In addition to these main effects, competence and relatedness showed an interaction effect with gender. Closer inspection of these interaction effects indicated that these two motivations had a significantly stronger effect on time spent playing if the player was male. In order to define why core players spent almost two hours every single day playing this game (*weekly mean* = 13.27, *SD* = 10.45), a linear regression model with all SDT motivations and gender as predictors of the weekly time spent playing was run F (4, 1609) 57.21, p <.001, $R^2 = 0.13$. Again, the need for relatedness was the strongest predictor ($b = 0.25$, $t = 11.00$, p <.001), followed by the need for competence ($b = 0.14$, $t = 5.81$, p <.001), male gender ($b = 0.08$, $t = 3.92$, p <.001), and the need for autonomy ($b = 0.06$, $t = 2.57$, p =.010).

4.4. Motivational moderation models of gender and microtransactions

The majority of core players (74.7 %) had spent money on Pokécoins for in-game purchases ($n = 1217$). Pokécoins were mostly used to purchase *Storage upgrade* (72.0 %), *Bag upgrades* (70.4 %), *Adventure Boxes* (66.6 %) and *Egg incubators* (55.5 %). We also found a weak positive association between age and spending (see Table 2) indicating that older players spent slightly more on microtransactions in Pokémon Go. Table 2 also shows weak positive associations between spending and the need for competence, and the need for relatedness. The motivational moderation models with gender are shown in Table 3. The need for competence remained a significant predictor of spending, whereas the need for relatedness fell below statistical significance ($t = 1.89$, p =.059). Instead, male gender was the significant predictor of spending in that model. The need for in-game autonomy had no effect on spending, and no interaction effects were observed. The sum score of gaming disorder symptoms did show a significant correlation with the amount that was spent on microtransactions ($r = 0.19$, p <.001). To determine which factors were the strongest predictors of player spending, we ran a linear regression model with all SDT motivations, age

Table 2
Zero-order correlations between motivations, player characteristics and usage.

	Experience Points	Pokémon Caught	Hours per Week	Spending
Age	0.25**	0.21**	0.05	0.05*
Competence	0.09**	0.11**	0.20**	0.17**
Autonomy	−0.03	−0.05	0.12**	0.02
Relatedness	0.38**	0.30**	0.31**	0.24**
Social Anxiety	−0.09**	−0.06*	0.05*	−0.01
IGD	0.22**	0.25**	0.31**	0.19**

Note: ** $p < .001$, * $p < .05$.

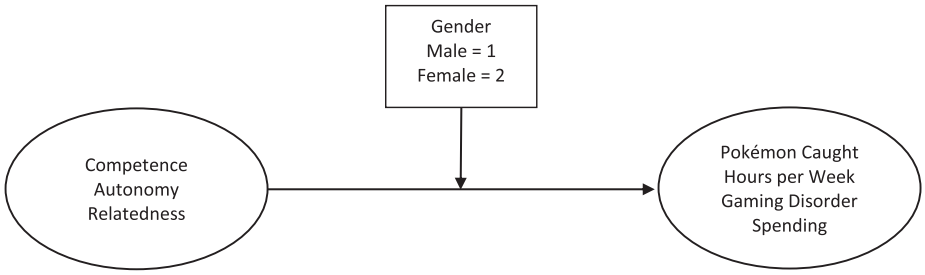


Fig. 1. The Moderation of Gender within the Effects of Motivations.

Table 3
Direct effects, and moderation effects of motivations and gender on selected types of Pokémon Go use.

Dependent Variable	Independent Variable	Independent Effect		Gender Effect *(male = 1, female = 2)*		Interaction Gender × Independent		Total Moderation Model
Pokémon caught	Competence	β = −0.93, SE = 1.85 $t = -0.50$, $p = .614$		β = −12.71, SE = 3.83 $t = -3.32$, $p = .001$	***	β = 1.97, SE = 1.11 $t = 1.77$, $p = .076$		$R = 0.21$, $R^2 = 0.04$, $F(3,1596) = 24.48$, $p < .001$
Pokémon caught	Autonomy	β = 0.19, SE = 1.57 $t = 0.12$, $p = .906$		β = −5.30, SE = 2.29 $t = -2.31$, $p = .021$	*	β = −0.47, SE = 0.94 $t = -0.50$, $p = .617$		$R = 0.18$, $R^2 = 0.03$, $F(3,1600) = 18.42$, $p < .001$
Pokémon caught	Relatedness	β = 6.38, SE = 1.48 $t = 4.31$, $p < .001$	***	β = −2.77, SE = 3.01 $t = -0.92$, $p = .357$		β = −0.83, SE = 0.86 $t = -0.97$, $p = .332$		$R = 0.34$, $R^2 = 0.12$, $F(3,1594) = 69.85$, $p < .001$
Hours per week	Competence	β = 4.66, SE = 1.09 $t = 4.27$, $p < .001$	***	β = 2.45, SE = 2.26 $t = 1.09$, $p = .278$		β = −1.32, SE = 0.66 $t = -2.00$, $p = .045$	*	$R = 0.23$, $R^2 = 0.05$, $F(3,1608) = 29.13$, $p < .001$
Hours per week	Autonomy	β = 2.26, SE = 0.94 $t = 2.39$, $p = .017$	*	β = −1.41, SE = 1.37 $t = -1.03$, $p = .303$		β = −0.50, SE = 0.56 $t = -0.89$, $p = .375$		$R = 0.17$, $R^2 = 0.03$, $F(3,1612) = 15.94$, $p < .001$
Hours per week	Relatedness	β = 4.92, SE = 0.89 $t = 5.54$, $p < .001$	***	β = 1.93, SE = 1.80 $t = 1.07$, $p = .284$		β = −1.11, SE = 0.52 $t = -2.13$, $p = .031$	*	$R = 0.32$, $R^2 = 0.10$, $F(3,1606) = 61.30$, $p < .001$
Spending	Competence	β = 0.20, SE = 0.08 $t = 2.38$, $p = .017$	*	β = −0.11, SE = 0.17 $t = 0.66$, $p = .512$		β = −0.03, SE = 0.05 $t = -0.57$, $p = .566$		$R = 0.21$, $R^2 = 0.04$, $F(3,1607) = 24.35$, $p < .001$
Spending	Autonomy	β = 0.07, SE = 0.07 $t = 0.94$, $p = .348$		β = −0.18, SE = 0.10 $t = -1.73$, $p = .090$		β = −0.03, SE = 0.04 $t = -0.61$, $p = .542$		$R = 0.15$, $R^2 = 0.02$, $F(3,1611) = 12.14$, $p < .001$
Spending	Relatedness	β = 0.13, SE = 0.07 $t = 1.89$, $p = .059$		β = −0.30, SE = 0.14 $t = -2.16$, $p = .031$	*	β = −0.03, SE = 0.04 $t = 0.75$, $p = .456$		$R = 0.27$, $R^2 = 0.07$, $F(3,1605) = 41.72$, $p < .001$
Gaming Disorder	Competence	β = 0.76, SE = 0.20 $t = 3.82$, $p < .001$	***	β = 0.29, SE = 0.41 $t = 0.72$, $p = .475$		β = −0.18, SE = 0.41 $t = -1.51$, $p = .131$		$R = 0.23$, $R^2 = 0.05$, $F(3,1504) = 28.18$, $p < .001$
Gaming Disorder	Autonomy	β = 0.40, SE = 0.17 $t = 2.35$, $p = .019$	*	β = 0.30, SE = 0.25 $t = -1.20$, $p = .231$		β = −0.06, SE = 0.10 $t = -0.54$, $p = .587$		$R = 0.19$, $R^2 = 0.04$, $F(3,1514) = 18.39$, $p < .001$
Gaming Disorder	Relatedness	β = 0.32, SE = 0.17 $t = 1.95$, $p = .052$		β = −0.31, SE = 0.33 $t = -0.92$, $p = .357$		β = −0.01, SE = 0.10 $t = -0.07$, $p = .945$		$R = 0.20$, $R^2 = 0.04$, $F(3,1514) = 20.92$, $p < .001$

Note: Pokémon caught β-values are x1000, *** p <.001, * p <.05.

and gender as predictors of the three categorizations of spending, F (5, 1607) 31.95, $p < .001$, $R^2 = 0.09$. Again, the need for relatedness was the strongest predictor ($b = 0.21$, $t = 8.65$, $p < .001$), followed by male gender ($b = 0.07$, $t = 3.00$, $p = .003$), the need for competence ($b = 0.12$, $t = 4.85$, $p < .001$), and age ($b = 0.08$, $t = 3.92$, $p < .001$). The need for autonomy did not predict spending on Pokémon Go ($b = -0.02$, $t = -0.78$, $p = .438$).

4.5. Gaming disorder, gender and social anxiety

To understand to what extend players of Pokémon Go show signs of gaming disorder, both DSM-5 and ICD-11 measures were applied (n = 1531). The DSM-5 describes that the threshold for IGD is experiencing five (or more) of the nine criteria (*Preoccupation, Tolerance, Withdrawal, Persistence, Escapism, Problems, Displacement, Deception, Conflict*) during a period of 12 months. Using the proposed 5-item cut-off point, 134 out of 1531 respondents who completed the IGD scale (8.8 %) met the tentative diagnosis for Internet Gaming Disorder [2]. Male players were significantly more likely to be meet this diagnostic threshold (11.4 %, n = 66) than female players (7.1 %, n = 67, $x^2(1) = 8.26$, p =.005). The WHO describes Gaming Disorder as pattern of persistent or recurrent gaming behavior that is excessive, obsessive and problematic. A relatively small group of 15 players (1.0 %) met the diagnosis for Gaming Disorder (WHO, 2019) by reporting excessive, obsessive, and problematic Pokémon Go use. With this measure there was no significant difference between players with gaming disorder between male players (1.4 %) and female players (0.7 %), $x^2(1) = 1.51$, p =.219). Although the sum scores for the two diagnostic measures show a strong correlation (r = 0.73, p <.001), the frequencies of players with gaming disorder using the IGD diagnostic cut-off (8.8 %) was almost nine times the size of those who met the three GD criteria (1 %). Table 4 further indicates that the tools for diagnosing players with gaming disorder are similar, showing near-identical mean scores on Pokémon caught, hours spent playing, spending and social anxiety. However, for the GD measure these were not always significantly different from non-disordered players, most likely due to statistical power deficiency related to the size of that sample (n = 15).

Similar to the previous tests, PROCESS was used to test whether gender moderated the effects of in-game motivational needs on the sum score of gaming disorder symptoms. As Table 3 shows, all three motivational needs significantly predicted IGD symptoms, negating earlier gender differences in IGD (see Table 1). Finally, we tested a mediation model in PROCESS to test if there was an indirect effect of social anxiety via player motivations (competence, autonomy, and relatedness) on the reported sum of criteria for IGD. The model is shown in Fig. 2, including corresponding standardized effect sizes. This model provided a significant prediction: model $R = 0.21$, $R^2 = 0.05$, $F(4,1526) = 76.41$, p <.001. Social anxiety had a significant total effect on the sum of IGD criteria (β = 0.50, SE = 0.06, t = 8.74, p <.001, LLCI 0.3853, ULCI 0.6082). Social anxiety also significantly influenced the motivational need for autonomy (β = 0.21, SE = 0.03, t = 7.29, p <.001), and showed a negative effect on the need for in-game relatedness (β = −0.10, SE = 0.03, t = −2.95, p =.003). Social anxiety did not influence the need for competence (β = 0.02, SE = 0.03, t = 0.61, p =.540). The mediating motivations all showed significant effects on the sum of IGD criteria: competence (β = 0.40, SE = 0.06, t = 7.00, p <.001), autonomy (β = 0.12, SE = 0.05, t = 2.36, p =.018), and relatedness (β = 0.27, SE = 0.04, t = 6.03, p <.001). In addition, there was a direct effect of social anxiety on IGD symptoms (β = 0.48, SE = 0.06, t = 8.61, p <.001).

Table 4
Similarities between IGD and GD diagnostic tools.

	Internet Gaming Disorder (DSM-5) (n = 134)	Neutral (n = 1397)	Gaming Disorder (ICD-11) (n = 15)	Neutral (n = 1516)
Pokémon caught	(*M* 33.58, *SD* 17.78) - (*M* 24.12, *SD* 17.07) *t* (1518) −6.11, *p* <.001		(*M* 32.50, *SD* 17.93) - (*M* 24.88. *SD* 17.32) *t* (1518) −1.70, *p* =.090	
Hours per week	(*M* 21.13, *SD* 16.34) - (*M* 12.37. *SD* 9.22) *t* (141.45) −6.19, *p* <.001		(*M* 22.83, *SD* 16.18) - (*M* 13.04. *SD* 10.20) *t* (14.11) −2.34, *p* =.035	
Spending	(*M* 2.42, *SD* 0.73) - (*M* 2.10. *SD* 0.78) *t* (1518) −4.59, *p* <.001		(*M* 2.50, *SD* 0.71) - (*M* 2.11. *SD* 0.78) *t* (1528) −1.38, *p* =.169	
Social anxiety	(*M* 2.42, *SD* 0.73) - (*M* 1.79. *SD* 0.77) *t* (148.38) −5.17, *p* <.001		(*M* 2.40, *SD* 0.82) - (*M* 2.12. *SD* 0.78) *t* (14.08) −2.43, *p* =.029	

5. Discussion

The aim of this study was to examine the characteristics, motivations and spending behavior of core Pokémon Go players by analyzing self-reported data from 1630 questionnaires. The motives for playing were examined using the Self-Determination Theory [11]. This theory proposes that the need for competence, autonomy and relatedness are innate universal psychological needs that motivate individuals to initiate behavior [11]. Results indicated that the need for in-game relatedness (i.e., a sense of community and friendship) was generally the most important motivational need among core players. This need to experience meaningful interaction with other players was the strongest predictor of the number of Pokémon caught, the weekly time spent playing, and the amount of money spent on microtransactions. The importance of social interaction in Pokémon Go on continued play has been shown in several other studies (e.g., [7,14]; [54]). Additionally, the interaction with gender indicated that the need for in-game relatedness driving time spent playing was strongest among male players, providing further evidence that male players show a stronger motivational need for relatedness [89]. Our results also confirmed the importance of experiencing in-game relatedness on making in-game purchases (e.g., [15]; [35]). Thus, meaningful in-game interactions with others and a strong sense of community are essential for the long-term popularity and financial success of Pokémon Go.

In line with previous studies, the need for competence (i.e., effective progress through the game) was very important to players [4,15], especially among male players. Contrary to a recent study however [35], competence was not the *most* important factor for playing the game among the current sample of core gamers. Although players with a stronger need for competence generally caught more than the sample-mean of twenty-five thousand Pokémon, the need for competence was a weak predictor of the franchise's motto '*Gotta catch 'em al*'. The moderation analysis indicated that male gender proved more important for catching Pokémon than the need for competence. This finding provides further evidence that male players generally show a stronger preference for competition in games [20,42]. Similar to previous findings [19], players who sought competence-related gratifications in the game, spent more money on in-game content. The most-purchased types of content (storage- and bag upgrades) enable players to collect more items and more Pokémon, highlighting the competence-related desire for effective progression without obstructions [90]. Gender moderation analyses indicated that need for in-game competence was the only motivation that showed a significant weak effect on purchase behavior. Thus, in-game competence is considered a stable motivation for core players of Pokémon Go that also contributes to its financial success.

Players' need for in-game autonomy showed some interesting, seemingly contradictory results. Autonomy was operationalized as the need for aesthetic customization options because these are directly linked to the freedom to make in-game choices without providing any competitive advantage. Customization options generally result in more enjoyment when playing online games [21,32], and more time spent playing Pokémon Go specifically [91]. Similarly, our study showed that the need for autonomy was positively related to time spent playing the game. However, the need for autonomy was negatively related to the total amount of Pokémon caught and experience gained. This indicates that those who care more about the customization of their avatar may care less about catching Pokémon and therefore gain less experience. In line with previous findings [92], male players were mainly motivated by the need for in-game relatedness and competence, whereas female

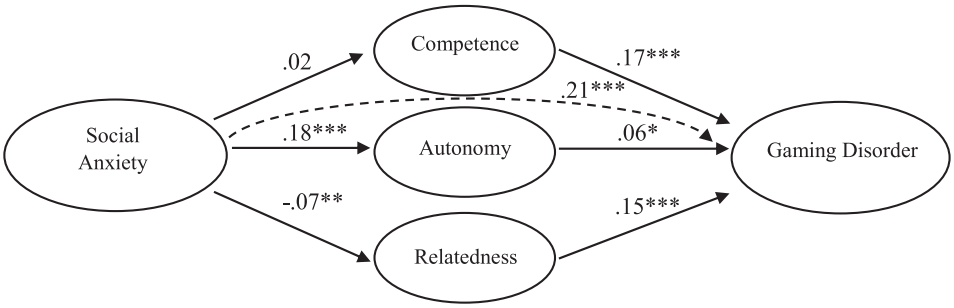


Fig. 2. Motivations within the Effect of Social Anxiety on Gaming Disorder Note: All coefficients are standardized effect sizes within the model, *p <.05, **p <.01, ***p <.001.

players were more motivated than men by the need to experience a sense of autonomy in the game. These motivational gender differences also follow observations by López-Fernández et al. [43] who found that male gamers scored higher on all motivations except the need for customization. Although several studies have shown that the need for customization may lead to spending because it can enhance a players' sense of authenticity [63] and identification [49], the current study did not find a relation between autonomy-related customization needs and spending on microtransactions. Similar to a recent study on Pokémon Go, players who were motivated by autonomy were not more likely to make in-game purchases [35]. As buying in-game skins has proven to be a very successful monetization technique in other free-to-play games [26], the current non-finding may be related to the lack of paid customization options available in Pokémon Go at the time of study,

In general, autonomy-related game elements such as customization options seem less important to core players who seek fulfillment of other needs through competence and relatedness aspects. Perhaps earlier studies on Pokémon Go failed to identify the need for in-game relatedness as a driving factor [4,15]because social elements such as the ability to form friendships with other trainers and the ability to trade Pokémon had not been implemented in the game yet. More recent studies, including the current study, find that a sense of social belonging, and community involvement are essential to players' long-term behavioral intentions to play the game [7,14,60]. Players seem primarily motivated by social interactions and contact with their community through trading and cooperating in raids. As other research has also shown, players who are motivated by initiating in-game relationships play the game more, find it more challenging, and find it more interesting [93]. Furthermore, because cooperation through raids and trading leads to more success in the game, it enhances players' sense of competence, reinforcing the effects of both motivations.

Contrary to the common understanding of males being hardcore gamers, the current sample of core players consisted of more female than male players. Core gamers are often identified based on their substantial investment of time and money on games [29]. Indeed, male players had started playing the game sooner after its release, had caught more Pokémon, had gathered more experience points, and had spent more money on the game than female players had. However, hardcore gamers have also been classified according to the types of games they play most (i.e., *First Person Shooters* and *Sports*). The group of casual gamers, that is dominated by female players, are known to predominantly play free online collaborative games such as Pokémon Go, typically for shorter periods than male players [46]. Regardless, most core players of Pokémon Go had spent money on this free game. Players with a higher behavioral investment (time spent per week, total Pokémon caught, and total xp gained) also showed higher financial investments. These findings emphasize the importance of player loyalty for purchasing virtual goods [17,24]. However, one important shortcoming in this study is the categorization of spending patterns. Since 37 % of the players indicated having spent more than 50 euros (the highest answer category), a wider range of spending options would have been appropriate.

5.1. Gaming disorder and Pokémon Go

Two measures of gaming disorder were applied to determine the prevalence of this tentative disorder in our sample of core players: a 9-item dichotomous IGD scale and a 3-item gaming disorder scale. The IGD scale [37] indicated that Internet gaming disorder [2] was present among almost 9 % of the players, whereas affirmative answers to the three items for Gaming disorder [62] indicated a prevalence rate of only 1 %. The discrepancy between these measures is likely related to the specific criteria required for a positive diagnosis. For IGD, the proposed diagnostic cut-off point of five criteria was reportedly conservatively chosen because a lower threshold would inflate diagnoses and could result in the classification of individuals who have not suffered significant clinical impairment [94]. However, this measure may still stigmatize highly engaged gamers, who may indeed experience signs of excessive and obsessive gaming, but fail to experience significant life impairment as a consequence of their gaming [10,95]. It seems pertinent that any measure of gaming disorder includes significant life impairment as a necessary criterion. The three requisite WHO-criteria, summarized as excessive, obsessive and problematic gaming patterns therefore seem to provide a better estimate of the prevalence of gaming disorder, although this measure is still far from ideal. The percentage of players with gaming disorder when assessed with the WHO criteria is less than the estimated worldwide prevalence of 2 % and 3 % [96]. Another shortcoming of the current study is that the measurement of gaming disorder consisted of three overlapping items from the IGD scale, likely inflating the correlation between these scales. Furthermore, prevalence rates lower than expected may seem somewhat dubious, considering the intense behavioral investment made by these core players. Further research is needed to determine the validity and reliability of gaming disorder measurements.

Regardless of the measure applied to determine whether players are experiencing gaming disorder, players with gaming disorder differ significantly from the overwhelming majority of core players that form the backbone of the Pokémon Go community. Players with gaming disorder may manifest maladaptive cognitions, including an over-reliance on gaming for self-worth and identity [97,98]. Indeed, disordered gamers may seek a sense of in-game autonomy, relatedness and competence to compensate these deficiencies in their daily lives. Furthermore, previous research on social anxiety suggested that Pokémon Go might be used as a social opportunity for those who are socially anxious, while simultaneously preventing these players from fully participating in all aspects of Pokémon Go [100]. Similar to previous findings on this game, socially anxious players tended to catch less Pokémon and gained less experience points during gameplay [30], possibly due to reduced social interaction and associated success that follows from raids on exceptional Pokémon. Social anxiety has also been shown to be an antecedent of game addiction [39,80]. Our mediation model with social anxiety as a predictor of gaming disorder, mediated by the three motivational needs, indicated that social anxiety among players predicts signs of gaming disorder, both directly and indirectly

through their need for autonomy. The indirect effect also passed through a reduced need for relatedness on gaming disorder, accentuating the aversion of online contact among these players. From a social and clinical perspective it seems relevant for future research to examine which features of Pokémon Go have a positive influence on those who show signs of psychosocial vulnerabilities or gaming disorder in order to enhance well-being among this vulnerable group.

6. Conclusion

Although Pokémon Go has been a popular topic of research since its release in 2016, there are still many questions unanswered. The aim of this study was to examine the characteristics, motivations and spending behavior of core Pokémon Go players. Results from an online survey among 1630 core players showed that the most prominent motivation was the need for in-game relatedness. Male players felt a higher need for competence and relatedness than female players, whereas female players felt a higher need for autonomy. The majority of players had spent money on in-game items or abilities, most of which were related to the need for competence. Between 1 and 9 percent could be considered players with gaming disorder based on the diagnostic criteria from either ICD-11 [62] or DSM-5 [2]. These players with gaming disorder were more likely male, spent more money on microtransactions, and showed more signs of social anxiety. It is important to note the temporality of the current findings as many social- and customization features, a new level cap, as well as a couple of hundred new Pokémon, have been added to the game since the data was collected, As the game continues to develop, so too will its core players. The longer they have been playing, and the higher their frequency of play, the less likely they are to stop playing, especially if these players are socially active in the game [31].

This study provides empirical support for incorporating motivational factors, gender, psychosocial characteristics, and gaming disorder in order to understand the persistent playing habits of core gamers. This study also contributes to the literature on *Freemium* games by highlighting the importance of gaming disorder for in-game microtransactions, especially among competence and relatedness-seeking male players. Knowledge of the motivations of dedicated players allows game designers to create more effective game experiences, tailored to the needs of target audiences, but it can also lead to obsessive and excessive gaming habits that cannot be controlled despite the negative effects it may have on players' social-emotional well-being. The vast majority of players reported no negative outcome from playing. In fact, walking around, catching hundreds of Pokémon every week has been enjoyable enough to keep many players engaged for years. Yet the main motivation to keep playing comes from interactions with other players in the community. Among core players of Pokémon Go, catching them all seems less important than catching up with friends.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

[1] T. Althoff, R.W. White, E. Horvitz, Influence of Pokémon Go on physical activity: study and implications, J. Med. Internet Res. 18 (12) (2016) 315.
[2] APA, Diagnostic and Statistical Manual of Mental Disorders, 5th edition, American Psychiatric Association, Washington, DC, 2013.
[3] App Annie (2020). Retrieved from https://www.appannie.com/en/apps/ios/app/1094591345/.
[4] D.R. Broom, K.Y. Lee, M.H.S. Lam, S.W. Flint, Gotta catch 'em all or not enough time: Users motivations for playing Pokémon Go™ and non-users' reasons for not installing, Health Psychol. Res. 7 (2019).
[5] J.A. Bonus, A. Peebles, M.L. Mares, I.G. Sarmiento, Look on the bright side (of media effects): Pokémon Go as a catalyst for positive life experiences, Media Psychol. 21 (2) (2018) 263–287.
[6] G.A. Brooks, L. Clark, Associations between loot box use, problematic gaming and gambling, and gambling-related cognitions, Addict. Behav. 96 (2019) 26–34.
[7] S. Bueno, M.D. Gallego, J. Noyes, Uses and gratifications on augmented reality games: An examination of pokémon go, Appl. Sci. 10 (2020) 1644.
[8] A. Chen, Y. Lu, B. Wang, Enhancing perceived enjoyment in social games through social and gaming factors, Inf. Technol. People (2016).
[9] Y. Cheng, The mediating effects of motivation for playing Pokémon Go on internet gaming disorder and well–being, Am. J. Fam. Ther. (2019) 1–18.
[10] J.P. Charlton, I.D.W. Danforth, Distinguishing addiction and high engagement in the context of online game playing, Comput. Hum. Behav. 23 (2007) 1531–1548.
[11] E.L. Deci, R.M. Ryan, The "what" and "why" of goal pursuits: Human needs and the self-determination of behavior, Psychol. Inq. 11 (2000) 319–338.
[12] A.R. Dewar, K.A. Kapalo, M.A. Rupp, J.L. Szalma, Using self-determination theory to examine differences in motvation and engagement in video gamers, *in: Proceedings of the Human Factors and Ergonomics Society Annual Meeting 59 (*2015*)* 882-886.
[13] M. Dreier, K. Wölfling, E. Duven, S. Giralt, M.E. Beutel, K.W. Müller, Free-to-play: about addicted Whales, at risk Dolphins and healthy Minnows. Monetarization design and internet gaming disorder, Addict. Behav. 64 (2017) 328–333.
[14] P.J. Ewell, M.C. Quist, C.S. Øverup, H. Watkins, R.E. Guadagno, Catching more than pocket monsters: Pokémon Go's social and psychological effects on players, J. Soc. Psychol. 160 (2020) 131–136.
[15] E.M. Ghazali, D.S. Mutum, M.Y. Woon, Multiple sequential mediation in an extended uses and gratifications model of augmented reality game Pokémon Go, Internet Res. (2019).
[16] R. Green, P.H. Delfabbro, D.L. King, Avatar identification and problematic gaming: The role of self-concept clarity, Addict. Behav. 113 (2020), 106694.
[17] J. Hamari, Why do people buy virtual goods? Attitude toward virtual good purchases versus game enjoyment, Int. J. Inf. Manag. 35 (3) (2015) 299–308.
[18] J. Hamari, L. Keronen, Why do people buy virtual goods: A meta-analysis, Comput. Hum. Behav. 71 (2017) 59–69.
[19] J. Hamari, A. Malik, J. Koski, A. Johri, Uses and Gratifications of Pokémon Go: Why do People Play Mobile Location-Based Augmented Reality Games? Int. J. Human-Computer Interaction 35 (2018) 1–16.
[20] T. Hartmann, C. Klimmt, Gender and computer games: Exploring females' dislikes, J. Comput.-Mediat. Commun. 11 (4) (2006) 910–931.
[21] J. Hilgard, C.R. Engelhardt, B.D. Bartholow, Individual differences in motives, preferences, and pathology in video games: the gaming attitudes, motives, and experiences scales (GAMES), Front. Psychol. 4 (2013) 608.
[23] K.B. Howe, C. Suharlim, P. Ueda, D. Howe, I. Kawachi, E.B. Rimm, Gotta catch'em all! Pokémon GO and physical activity among young adults: difference in differences study, BMJ 355 (2016), i6270.
[24] K.L. Hsiao, C.C. Chen, What drives in-app purchase intention for mobile games? An examination of perceived values and loyalty, Electron. Commer. Res. Appl. 16 (2016) 18–29.
[25] E.E. Hussein, T.J. Shiang, A.I. Wehelie, Mediating effect of attitude on awareness toward the addiction behaviour of Pokémon Go players in Malaysia, J. Manage. Market. Rev. 2 (2017) 8–13.
[26] J. Jarrett, Gaming the gift: The affective economy of League of Legends 'fair'free-to-play model, J. Consum. Cult. 21 (2021) 102–119.
[27] N. Jenkins, *Pokémon Go May Have Just Shown Us What the End of the World Looks Like*, 2016, August 22. Retrieved from https://time.com/4460911/pokemon-go-taipei-stampede-snorlax-mob-xinbeitou-taiwan/.
[28] M. Jula, *Pokémon Go Players Take Their Hunt to the Streets of New York,* 2016, July 11. Retrieved from https://www.nytimes.com/2016/07/12/nyregion/pokemon-players-take-their-hunt-to-the-streets-of-new-york.html.
[29] J. Juul, A casual revolution: Reinventing video games and their players, MIT press, 2010.
[30] A. Khalis, A.Y. Mikami, Who's Gotta Catch 'Em All?: Individual differences in Pokèmon Go gameplay behaviors, Pers. Individ. Differ. 124 (2018) 35–38.
[31] S. Kim, D. Choi, E. Lee, W. Rhee, Churn prediction of mobile and online casual games using play log data, PLoS One 12 (2017) e0180735.
[32] K. Kim, M.G. Schmierbach, M.Y. Chung, J.D. Fraustino, F. Dardis, L. Ahern, Is it a sense of autonomy, control, or attachment? Exploring the effects of in-game customization on game enjoyment, Comput. Hum. Behav. 48 (2015) 695–705.
[33] D.L. King, P.H. Delfabbro, Predatory monetization schemes in video games (eg 'loot boxes') and internet gaming disorder, Addiction 113 (2018) 1967–1969.
[34] D.L. King, A.M. Russell, P.H. Delfabbro, D. Polisena, Fortnite microtransaction spending was associated with peers' purchasing behaviors but not gaming disorder symptoms, Addict. Behav. 104 (2020), 106311.
[35] M. Kosa, A. Uysal, The role of need satisfaction in explaining intentions to purchase and play in Pokémon Go and the moderating role of prior experience, Psychol. Popular Media 10 (2021) 187–200.
[37] J.S. Lemmens, P.M. Valkenburg, D.A. Gentile, The Internet gaming disorder scale, Psychol. Assess. 27 (2) (2015) 567.
[38] J.S. Lemmens, P.M. Valkenburg, J. Peter, Development and validation of a game addiction scale for adolescents, Media Psychol. 12 (2009) 77–95.
[39] J.S. Lemmens, P.M. Valkenburg, J. Peter, Psychosocial causes and consequences of pathological gaming, Comput. Hum. Behav. 27 (1) (2011) 144–152.
[40] C. Levesque, A.N. Zuehlke, L.R. Stanek, R.M. Ryan, Autonomy and competence in German and American university students: A comparative study based on self-determination theory, J. Educ. Psychol. 96 (2004) 68.
[41] H. Li, Y. Liu, X. Xu, J. Heikkilä, H. Heijden, Modeling hedonic is continuance through the uses and gratifications theory: An empirical study in online games, Comput. Human Behavior 48 (2015) 261–272.

[42] K. Lucas, J.L. Sherry, Sex differences in video game play: A communication-based explanation, Commun. Res. 31 (5) (2004) 499–523.
[43] F.J. López-Fernández, L. Mezquita, M.D. Griffiths, G. Ortet, M.I. Ibáñez, The development and validation of the Videogaming Motives Questionnaire (VMQ), PLoS One 15 (10) (2020) e0240726.
[44] A. Malik, K. Hiekkanen, Z. Hussain, J. Hamari, A. Johri, How players across gender and age experience Pokémon Go? Univ. Access Inf. Soc. 19 (4) (2020) 799–812.
[45] O. Marquet, C. Alberico, D. Adlakha, J.A. Hipp, Examining motivations to play Pokémon GO and their influence on perceived outcomes and physical activity, JMIR Serious Games 5 (2017) 21.
[46] L. McLean, M.D. Griffiths, Female gamers' experience of online harassment and social support in online gaming: A qualitative study, Int. J. Ment. Heal. Addict. 17 (2019) 970–994.
[47] D.J. Mills, M. Milyavskaya, J. Mettler, N.L. Heath, Exploring the pull and push underlying problem video game use: A Self-Determination Theory approach, Pers. Individ. Differ. 135 (2018) 176–181.
[48] Newzoo, 2019. Retrieved from https://newzoo.com/insights/trend-reports/newzoo-global-games-market-report-2019-light-version/.
[49] B.W. Park, K.C. Lee, Exploring the value of purchasing online game items, Comput. Hum. Behav. 27 (2011) 2178–2185.
[50] W. Peng, J.H. Lin, K.A. Pfeiffer, B. Winn, Need satisfaction supportive game features as motivational determinants: An experimental study of a self-determination theory guided exergame, Media Psychol. 15 (2012) 175–196.
[52] A. Pourmand, K. Lombardi, E. Kuhl, F. O'Connell, Videogame-related illness and injury: a review of the literature and predictions for Pokemon GO!, Games Health J. 6 (1) (2017) 9–18.
[53] A.K. Przybylski, C.S. Rigby, R.M. Ryan, A motivational model of video game engagement, Rev. Gen. Psychol. 14 (2010) 154–166.
[54] P.A. Rauschnabel, A. Rossmann, M.C. tom Dieck, An adoption framework for mobile augmented reality games: The case of Pokémon Go, Comput. Hum. Behav. 76 (2017) 276–286.
[55] P.C. Raneri, C. Montag, D. Rozgonjuk, J. Satel, H.M. Pontes, The role of microtransactions in internet gaming disorder and gambling disorder; A preregistered systematic review, Addict. Behav. Rep. 15 (2022) 1–16.
[56] A.M. Rubin, Uses and gratifications, in: The SAGE Handbook of Media Processes and Effects, SAGE, Los Angeles, 2009, pp. 147–159.
[57] R.M. Ryan, C.S. Rigby, A. Przybylski, The motivational pull of video games: A self-determination theory approach, Motiv. Emot. 30 (4) (2006) 344–360.
[58] M. Scharkow, R. Festl, J. Vogelgesang, T. Quandt, Beyond the "core-gamer": Genre preferences and gratifications in computer games, Comput. Hum. Behav. 44 (2015) 293–298.
[59] C.I. Teng, How avatars create identification and loyalty among online gamers, Internet Res. 29 (2019) 1443–1468.
[60] K. Vella, D. Johnson, V.W.S. Cheng, T. Davenport, J. Mitchell, M. Klarkowski, C. Phillips, A sense of belonging: Pokemon GO and Social Connectedness, Games Culture 14 (6) (2019) 583–603.
[61] A.I. Wang, Systematic literature review on health effects of playing Pokémon Go, Entertainment Comput. 38 (2021), 100411.
[62] World Health Organization, International classification of diseases for mortality and morbidity statistics (11th Revision), 2018. Retrieved from https://icd.who.int/browse11/l-m/en.
[63] S.L. Wu, C.P. Hsu, Role of authenticity in massively multiplayer online role playing games (MMORPGs): Determinants of virtual item purchase intention, J. Bus. Res. 92 (2018) 242–249.
[64] F.J. Zach, I.P. Tussyadiah, in: To Catch Them All—the (Un) Intended Consequences of Pokémon GO on Mobility, Consumption, and Wellbeing, Springer, Cham, 2017, pp. 217–227.
[66] R. Tamborini, M. Grizzard, N.D. Bowman, L. Reinecke, R.J. Lewis, A. Eden, Media enjoyment as need satisfaction: The contribution of hedonic and nonhedonic needs, Journal of Communication 61 (6) (2011) 1025–1042.
[67] C. Burgers, A. Eden, M.D. van Engelenburg, S. Buningh, How feedback boosts motivation and play in a brain-training game, Computers in Human Behavior 48 (2015) 94–103.
[68] C. Burgers, A. Eden, M.D. van Engelenburg, S. Buningh, How feedback boosts motivation and play in a brain-training game, Computers in Human Behavior 48 (2015) 94–103.
[69] H.J. Liu, Y.C. Shiue, Influence of Facebook game players' behavior on flow and purchase intention, Social Behavior and Personality: an international journal 42 (1) (2014) 125–133.
[70] D.R. Broom, S.W. Flint, Gotta catch 'em all: Impact of Pokémon Go on physical activity, sitting time, and perceptions of physical activity and health at baseline and three-month follow-up, Games for health journal 7 (6) (2018) 401–408.
[71] J. Evans, S.Z. Evans, D.B. Shank, Q.P. Fallon, Motivations for social interaction: The case of Pokémon go after the fad ended, Social Science Quarterly 102 (1) (2021) 547–551.
[73] S.A. Nikou, A.A. Economides, Mobile-Based Assessment: Integrating acceptance and motivational factors into a combined model of Self-Determination Theory and Technology Acceptance, Computers in Human Behavior 68 (2017) 83–95.
[74] C. Burgers, A. Eden, M.D. van Engelenburg, S. Buningh, How feedback boosts motivation and play in a brain-training game, Computers in Human Behavior 48 (2015) 94–103.
[75] Y. Qin, Attractiveness of game elements, presence, and enjoyment of mobile augmented reality games: The case of Pokemon Go, Telematics and Informatics 62 (2021), 101620.
[76] K. Alha, E. Koskinen, J. Paavilainen, J. Hamari, Why do people play location-based augmented reality games: a study on Pokémon GO, Computers in Human Behavior 93 (2019) 114–122.
[77] L. Hjorth, I. Richardson, Pokémon GO: Mobile media play, place-making, and the digital wayfarer, Mobile Media Communication 5 (1) (2017) 3–14.
[78] D.L. King, P.H. Delfabbro, The cognitive psychopathology of Internet gaming disorder in adolescence, Journal of abnormal child psychology 44 (8) (2016) 1635–1645.
[80] D.L. King, P.H. Delfabbro, J.C. Perales, J. Deleuze, O. Király, E. Krossbakken, J. Billieux, Maladaptive player-game relationships in problematic gaming and gaming disorder: A systematic review, Clinical psychology review 73 (2019), 101777.
[81] S. Voigt, O. Hinz, Making digital freemium business models a success: Predicting customers' lifetime value via initial purchase information, Business Information Systems Engineering 58 (2) (2016) 107–118.
[82] D. Zendle, R. Meyer, N. Ballou, The changing face of desktop video game monetisation: An exploration of exposure to loot boxes, pay to win, and cosmetic microtransactions in the most-played Steam games of 2010–2019, PloS one 15 (5) (2020), e0232780.
[83] E. Ghazali, D.S. Mutum, M.Y. Woon, Exploring player behavior and motivations to continue playing Pokémon GO, Information Technology People, 2018.
[84] K.L. Hsiao, M.D. Lytras, C.C. Chen, An in-app purchase framework for location-based AR games: the case of Pokémon Go, Library Hi Tech (2019).
[85] World Health Organization. (2019). ICD-11: International classification of diseases (11th revision). Retrieved from https://icd.who.int/.
[86] R.P. Mattick, J.C. Clarke, Development and validation of measures of social phobia scrutiny fear and social interaction anxiety, Behaviour Research and Therapy 36 (4) (1998) 455–470.
[87] M. Dreier, K. Wölfling, E. Duven, S. Giralt, M.E. Beutel, K.W. Müller, Free-to-play: about addicted Whales, at risk Dolphins and healthy Minnows. Monetarization design and internet gaming disorder, Addict. Behav. 64 (2017) 328–333.
[89] C.J. Ferguson, C.K. Olson, Friends, fun, frustration and fantasy: Child motivations for video game play, Motivation and Emotion 37 (1) (2013) 154–164.
[90] J. Hamari, K. Alha, S. Järvelä, J.M. Kivikangas, J. Koivisto, J. Paavilainen, Why do players buy in-game content? An empirical study on concrete purchase motivations, Computers in Human Behavior 68 (2017) 538–546.
[91] Y. Qin, Attractiveness of game elements, presence, and enjoyment of mobile augmented reality games: The case of Pokemon Go, Telematics and Informatics 62 (2021), 101620.
[92] A. Malik, K. Hiekkanen, Z. Hussain, J. Hamari, A. Johri, How players across gender and age experience Pokémon Go? Universal Access in the Information Society 19 (4) (2020) 799–812.
[93] J. Evans, S.Z. Evans, D.B. Shank, Q.P. Fallon, Motivations for social interaction: The case of Pokémon go after the fad ended, Social Science Quarterly 102 (1) (2021) 547–551.
[94] N.M. Petry, F. Rehbein, D.A. Gentile, J.S. Lemmens, H.J. Rumpf, T. Mossle, et al., An international consensus for assessing internet gaming disorder using the new DSM-5 approach, Addiction. 109 (2014) 1399–1406.
[95] A.J. Van Rooij, C.J. Ferguson, M. Colder Carras, D. Kardefelt-Winther, J. Shi, E. Aarseth, et al., A weak scientific basis for gaming disorder: Let us err on the side of caution, Journal of Behavioral Addictions 7 (1) (2018) 1–9.
[96] M.W. Stevens, D. Dorstyn, P.H. Delfabbro, D.L. King, Global prevalence of gaming disorder: A systematic review and meta-analysis, Australian New Zealand Journal of Psychiatry 55 (6) (2021) 553–568.
[97] S.E. Allison, L. Von Wahlde, T. Shockley, G.O. Gabbard, The development of the self in the era of the internet and role-playing fantasy games, American Journal of Psychiatry 163 (3) (2006) 381–385.
[98] C.L. Beard, R.E. Wickham, Gaming-contingent self-worth, gaming motivation, and internet gaming disorder, Computers in Human Behavior 61 (2016) 507–515.
[99] D.L. King, P.H. Delfabbro, The cognitive psychopathology of Internet gaming disorder in adolescence, Journal of abnormal child psychology 44 (8) (2016) 1635–1645.
[100] T.A. Kato, A.R. Teo, M. Tateno, M. Watabe, H. Kubo, S. Kanba, Can Pokémon GO rescue shut-ins (hikikomori) from their isolated world, Psychiatry Clin Neurosci 71 (1) (2017) 75–76.
[101] D.S. Mennin, D.M. Fresco, R.G. Heimberg, F.R. Schneier, S.O. Davies, M. R. Liebowitz, Screening for social anxiety disorder in the clinical setting: using the Liebowitz Social Anxiety Scale, Journal of anxiety disorders 16 (6) (2002) 661–673.
[102] C.Z. Liu, Y.A. Au, H.S. Choi, Effects of freemium strategy in the mobile app market: An empirical study of google play, Journal of Management Information Systems 31 (3) (2014) 326–354.